

Scientific Note & Methodology

GreenRoute AI

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Abstract

This document serves as the official **Scientific Note** for GreenRoute AI. It details the precise datasets utilized, mathematical formulations of green metrics, biases and limitations of the machine learning models, and the strict Human-in-the-Loop (HITL) protocol implemented to ensure scientific transparency and full compliance with the hackathon’s rules.

1 1. Data Sources & Traceability: The Tox21 Dataset

To ensure complete traceability, avoid "pseudo-science," and guarantee highly accurate toxicity predictions, GreenRoute AI exclusively utilizes the **Tox21 Data Challenge dataset**.

1.1 Scientific Justification for Tox21

The Tox21 dataset is a premier resource generated by a collaborative initiative including the NIH, EPA, and FDA, measuring approximately 12,000 environmental chemicals and drugs across 12 different toxic effects via in vitro high-throughput screening. These effects cover critical human health pathways, including 7 nuclear receptor (NR) and 5 stress response (SR) pathways. We rely on the foundational **DeepTox** methodology (Mayr et al., 2016). DeepTox demonstrated that deep learning architectures naturally enable multi-task learning across all 12 assays, successfully identifying abstract chemical features (toxicophores). Because Tox21 provides experimentally validated data on highly diverse pathways, it is comprehensively sufficient for our green solvent screening. We did not require supplementary datasets, as Tox21 allows our models to reliably predict endocrine disruption, liver damage, and general cytotoxicity without the noise introduced by smaller, less standardized databases.

2 2. Green Metrics & Mathematical Formulations

The analytical scoring module ranks solvents based on an explicit, mathematically defined objective function.

2.1 Green Solvent Score (Configurable by User)

The overall Green Score is calculated as a weighted sum of normalized variables:

$$\text{Score} = 0.3(1 - \tilde{T}) + 0.2(1 - \tilde{V}) + 0.3(B) + 0.2(R) \quad (1)$$

Where $\tilde{T}$ is normalized toxicity, $\tilde{V}$ is normalized VOC emissions, B is biodegradability, and R is recyclability.

Important System Capability: The formula above represents the **standard, default weighting** in the platform. However, GreenRoute AI grants researchers full control to tweak these distinct coefficients via the UI parameters. Depending on the synthetic priorities (e.g., strict zero-toxicity vs. high necessity for recycling), the chemist can dynamically adjust the importance of each metric, reflecting true Human-in-the-Loop design.

2.2 E-factor and Atom Economy

$$E\text{-factor} = \frac{\text{Total Waste (kg)}}{\text{Desired Product (kg)}} \quad (2)$$

$$\text{Atom Economy} = \left(\frac{\text{Molecular Weight of Desired Product}}{\sum \text{Molecular Weight of all Reactants}} \right) \times 100\% \quad (3)$$

3 3. Model Architecture & Explainability (XAI)

Our toxicity and yield prediction rely on a Random Forest regressor/classifier utilizing Morgan fingerprints generated via RDKit. To quantify uncertainty and avoid black-box assertions, we implement 50 iterations of Monte Carlo Dropout. This provides the chemist with a 95% confidence interval (e.g., 73.41%±6.34%) rather than a deterministic absolute.

4 4. Biases and Limitations

- **Imbalanced Data in Toxicity:** Toxicity datasets, including Tox21, inherently contain far more negative (non-toxic) than positive (toxic) samples. While our models use balancing techniques, rare toxicophores might still be under-predicted.
- **Descriptor Limits:** 2D descriptors do not fully capture 3D stereochemical constraints which may impact synthetic feasibility.

5 5. Human-in-the-Loop (HITL) Workflow

In compliance with Rule 4.2, GreenRoute AI acts purely as an advisory oracle. The workflow explicitly halts until the researcher tweaks their parameters and interacts with the "Approve for Experiment" module. No automated experimental execution can occur without human approval.

6 Appendix: Standard Reaction Validation Set

The following 5 common chemical reactions constitute the benchmark testing set used to validate the chemical plausibility of the AI’s solvent substitution recommendations:

1. **Amide Coupling (e.g., HATU-mediated):** Traditionally requires hazardous DMF or DCM. Evaluates the AI’s ability to suggest greener alternatives like 2-MeTHF or EtOAc.
2. **Suzuki-Miyaura Cross-Coupling:** Evaluates transition-metal catalysis solvent substitution (e.g., replacing Toluene with aqueous/surfactant conditions).
3. **Nucleophilic Aromatic Substitution (S_NAr):** Traditionally run in reprotoxic NMP or DMF. Used to test if the model recommends Cyrene or sulfolane.
4. **Grignard Formation/Addition:** Requires coordinating ethers. Evaluates the model’s capacity to favor 2-MeTHF over highly volatile THF or Diethyl Ether.
5. **Fischer Esterification:** Evaluates substitution of aromatics (Toluene/Benzene used for azeotropic water removal) with greener continuous removal methods or less toxic solvents.